Technische Universität München
Department of Aerospace and Geodesy
Professorship of Sustainable Future Mobility
Prof. Dr.-Ing. Agnes Jocher

Master's Thesis

Data Consistency Evaluation for Reaction Model Optimization

Author: Barkin Güney
Supervisor: Prof. Dr.-Ing. Agnes Jocher
Advisor: Dr. Nadezda Slavinskaya
Issue date: September 1, 2025
Submission date: March 02, 2026

Acknowledgements

Thank somebody if appropriate.

Abstract

Kurzzusammenfassung

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2.1 Background

2.1.1 Nonlinear Least Squares

2.2 Chemical Kinetics Database

TUMKIN, a chemical kinetic data infrastructure currently under development aims to incorporate databases and tools to facilitate analysis and optimization of reaction mechanisms of wide range of fuels. The idea is based on Process Informatics Model (PrIMe) created by Frenklach and coworkers which was a data infrastructure concentrating on data management, data uncertainty, and data consistency. [1] However PrIMe website is not active. TUMKIN aims to revive and expand on PrIMe. A similar effort is RESPECTH [2]. Their database contains direct and indirect experimental measurements for various fuels. Relevant for this thesis are the indirect shock tube ignition delay time measurements for Ethylene, Syngas and Hydrogen. [3][4][5] They also share computer programs for various tasks around analysis and optimization of kinetics. Reaction Mechanism Generator (RMG) is an automatic chemical reaction mechanism generator that constructs kinetic models composed of elementary chemical reaction steps using a general understanding of how molecules react. RMG database contain kinetics information for many reactions from various sources and datasets. [6]

2.3 Uncertainty Quantification in Combustion

Uncertainty quantification (UQ) is the science for quantification and minimization of uncertainties with computer experiments and simulation. [7] The approach to developing detailed kinetic models (mechanisms) of fuel combustion involves compiling a set of elementary reactions whose rate parameters may be determined from individual rate measurements, reaction-rate theory, or a combination of both. These methods have uncertainties. A widely used method is sensitivity analysis. UQ is closely related to sensitivity analysis, and in many ways, UQ is a natural extension of sensitivity analysis. A basic objective of UQ is to analyze the uncertainty in the model response because of model parameters being uncertain, including forward uncertainty quantification and propagation and an inverse problem leading to model uncertainty constraining. When treated as a Bayesian inference problem, UQ aids the development of predictive kinetic models by constraining models with data and yielding estimates for the confidence of a model to make predictions outside of the thermodynamic regimes where the model has been tested. [7]

2.4 Uncertainty of reaction mechanisms

Reaction mechanisms employed in chemical kinetics models include information about the relevant species, reactions, reaction rates and the corresponding uncertainties of these reaction rates. Reactions are modeled with the Arrhenius equation with k being the reaction rate coefficient (RRC). In the literature, Arrhenius coefficients (A,n,Ea) are fitted to experimental data with a regression model and evaluated with expert opinion. Or they are calculated from fundamental calculations. Therefore, different literature sources can report different reaction rate coefficient k(T) values for a given reaction. The uncertainty factor f is a logarithmic measure of the plausible spread of k(T) around its recommended value:

$$k(T) = AT^n \exp\left(-\frac{E_a}{T}\right) \qquad (\mathrm{cm}^3\,\mathrm{mol}^{-1}\,\mathrm{s}^{-1},\,\mathrm{K}) \tag{2.1}$$

To perform analysis or optimization of combustion mechanisms, it is important to quantify the uncertainty of its parameters. For every reaction there are many literature sources reporting varying parameter sets. The most comprehensive source of a database compiling such sources is NIST Kinetics Database. In this thesis, to quantify the uncertainty of Arrhenius expressions, literature sources are collected from RMG database. For every active reaction, nonlinear least squares fit is performed using the selected sources from the database. Fit is performed on the following regression model:

$$k(T) = 10^{\log_{10}(A)}\, T^n \exp\left(-\frac{E_a}{T}\right) \tag{2.2}$$

Parameter standard deviations $s\left(\log_{10}(A)\right), s(n), s(E_a)$ are used to calculate lower and upper uncertainty bounds for rate coefficients.

$$k_{\text{low}}(T) = 10^{\log_{10}(A) - s(\log_{10} A)}\, T^{n-s(n)} \exp\left(-\frac{E_a + s(E_a)}{T}\right) \tag{2.3}$$

$$k_{\mathrm{up}}(T) = 10^{\log_{10}(A)+s(\log_{10}A)}\, T^{n+s(n)} \, \exp\left(-\frac{E_a - s(E_a)}{T}\right) \tag{2.4}$$

The uncertainty parameter f is defined as:

$$f(T) = \log_{10}\left(\frac{k_{\mathrm{up}}(T)}{k_0(T)}\right) = \log_{10}\left(\frac{k_0(T)}{k_{\mathrm{low}}(T)}\right) \tag{2.5}$$

Where k0 is the best fit of the rate coefficient, klow and kup are the lower and the upper bounds respectively [8]. Rate coefficients are then sampled from the interval $\left[k_0\left(T\right)/10^f\left(T\right), k_0\left(T\right) * 10^f\left(T\right)\right]$. Which preserves the shape of the k(T) curve and only scales it. It is also possible to use the joint multivariate distribution of Arrhenius parameters (A,n,Ea) sample reaction rate coefficients. Covariance matrix However, authors rarely report covariance matrix or parameter standard deviations. When the covariance matrix is not available, often only preexponential factor A is varied to achieve the prescribed uncertainty band on $k\left(T\right)$ [9].

2.5 Sensitivity Analysis

Local sensitivity analysis is a method where the response of the model to small parameter changes close to nominal values is explored. Normalized first-order sensitivity coefficients are commonly applied in combustion to provide kinetic insight into the model and its strengths and weaknesses with respect to experimental data.[10]

2.6 Surrogate Models

In the B2BDC framework, surrogate models are constructed using a solution-mapping approach over the prior uncertainty region. Active parameters are first identified through impact-factor screening and the surrogate is built only in this reduced parameter space. Training data are generated using space-filling designs, and the original model is evaluated at each design point. Polynomial, quadratic, or rational quadratic surrogate models are then fitted by minimizing either $L_2 or L_\infty$ norms. Rational quadratic surrogates are used when polynomial models are insufficiently accurate, and their coefficients are obtained through semidefinite programming [11].

2.7 Consistency Analysis

2.7.1 MUM-PCE

In MUM-PCE, the uncertainty in the rate parameters of an as-compiled model is first quantified using stochastic spectral expansion and polynomial chaos techniques. The model is then constrained against global combustion experiments through solution mapping and least-squares optimization, combined with a consistency-screening procedure to identify incompatible experimental targets. The uncertainty of the constrained (posterior) model is subsequently calculated using an inverse spectral technique and propagated to a range of combustion conditions. [12]

2.7.2 B2BDC

Bound-to-Bound Data Collaboration (B2BDC) provides a natural framework for addressing both forward and inverse uncertainty quantification problems, in which QOI models are constrained by related experimental observations with interval uncertainty. B2BDC models uncertainty deterministically and the notion of validity is encapsulated in the consistency measure, and a collection of models, experimental observations, and parameter bounds determines a feasible region in the parameter space. If the feasible region is nonempty, the dataset is said to be consistent, whereas an empty feasible set indicates that models and data are fundamentally at odds. B2BDC is used to determine the consistency between a numerical model and an experimental dataset and to achieve reduction of the uncertainty, via inference from the data [13] [14]. A MATLAB implementation of B2BDC is provided by [11].

2.8 TUMKIN Implementations

2.8.1 Data Submission Protocol

Data submission protocol for experimental shock tube data is developed. Data should be given as described in template.json

UQ_barkin

2.8.2 Sensitivity Analysis

2.8.3 Consistency Analysis

Given a reaction mechanism, chemical kinetics simulation software like Canterа**cantera** can simulate combustion, and predict properties like ignition delay time(IDT), laminar flame speed(LFS), etc. These properties are also measured in real life combustion experiments, and reported together with quantified uncertainties. A good reaction model should be able to predict these properties accurately under same operating conditions as the physical experiment(temperature, pressure, stoichiometric ratio, mixture content). Consistency analysis in this context are methods to validate simulation and experimental results against each other by finding a suitable set of model parameters that best explain the experiment data within their uncertainty intervals. If some data can't be explained by to model, this can suggest either the data is bad, or the model is bad.

2.8.4 Possible Methods for consistency Analysis

In the example given in this B2BDC paper **Arun Hegde** Scalar consistency measure is enough to identify a feasible set for slightly inconsistent datasets, while a more detailed Vector consistency measure is necessary for highly inconsistent datasets. Since we are developing this consistency analysis to be implemented in TUMKIN, we want it to follow some practical considerations. - Work on different kinds of datasets.(different degrees of inconsistency, sensitivity analysis for large mecahnisms) - Take domain knowledge from user as needed. -

2.8.4.1 Bayesian History Matching

Implausibility metric

2.8.4.2 B2BDC Scalar Consistency Measure

Bound-to-Bound Data Collaboration (B2BDC) **Consistency Analysis for Massively Inconsistent Datasets in B**

Bound-to-Bound Data Collaboration (B2BDC) provides a natural framework for addressing both forward and inverse uncertainty quantification problems, in which quantity-of-interest (QOI) models are constrained by related experimental observations with interval uncertainty.

In this approach, uncertainty is modeled deterministically and the notion of validity is encapsulated in the consistency measure.

In B2BDC, models, experimental observations, and parameter bounds are taken as "tentatively entertained" and are combined into what is termed a dataset. A dataset determines a feasible region in the parameter space, defined as the set of parameter vectors for which models and data are in complete agreement within prescribed uncertainty bounds. If the feasible region is nonempty, the dataset is consistent, whereas if the feasible set is empty, then no parameter vector can lead to agreement between the models and the data.

A quantitative measure of agreement is provided by the scalar consistency measure, which characterizes this region by computing the maximal uniform constraint tightening associated with the dataset. The consistency measure is defined through an optimization problem that determines the largest simultaneous tightening of all experimental bounds for which a feasible parameter vector still exists. If the consistency measure is positive, all observation bounds can be simultaneously tightened without eliminating the feasible set. If the consistency measure is negative, the observation bounds are too restrictive and must be expanded in order for a feasible set to exist.

The B2BDC framework casts the problem of model validation in an optimization setting, where uncertainties are represented by intervals. Within a given physical model, parameters are constrained by the combination of prior knowledge and uncertainties in experimental data. A collection of such constraints constitutes a dataset and determines a feasible region in the parameter space. Assertions expressing additional knowledge, such as relationships among parameters or quantities of interest, can be incorporated as additional constraints.

B2BDC has been applied in several settings, including combustion science, atmospheric chemistry, quantum chemistry, system biology, and engineering design. Comparisons with Bayesian calibration approaches have shown that the two methods are similar in spirit and can yield overlapping predictions, while B2BDC provides conservative specifications and predictions when uncertainty information is limited.

In applied combustion modeling, B2BDC is used within a structured uncertainty-quantification workflow that includes quantification of experimental uncertainty, development and refinement of the physics model, identification of uncertain and sensitive parameters with prior bounds, sampling of the uncertain parameter space, and training of surrogate models. The final step, also referred to as the inverse problem, is performed by applying the Bound-to-Bound Data Collaboration approach to determine the consistency between a numerical model and an experimental dataset.

When datasets are inconsistent, B2BDC provides diagnostic tools to identify the sources of disagreement. Sensitivities of the consistency measure to perturbations in model-data constraints and parameter bounds are used to rank the degree to which individual constraints contribute to inconsistency. For datasets with numerous sources of inconsistency, an extension known as the vector consistency measure seeks the minimal number of independent constraint relaxations required to restore consistency.

Overall, B2BDC constitutes an optimization-based approach to model validation and uncertainty quantification, in which a model is considered predictive when its outcomes are accompanied by rigorously determined uncertainty bounds, and increased predictivity corresponds to more narrowly bounded interval estimations.

2.8.4.3 B2BDC Vector consistency Measure

2.8.4.4 F-Test

2.8.4.5 MUMPCE

Reliable simulations of reacting flow systems require a well-characterized, detailed chemical model as a foundation; however, the inherent uncertainties in the rate data leave a model characterized by a kinetic rate parameter space which will be persistently finite in its size. Without a careful analysis of how this uncertainty space propagates into the model predictions, those predictions can at best be trusted only semi-quantitatively. To address this issue, the Method of Uncertainty Minimization using Polynomial Chaos Expansions (MUM-PCE) is proposed to quantify and constrain these uncertainties.

In this method, the uncertainty in the rate parameters of the as-compiled model is quantified. The model is then subjected to a rigorous mathematical analysis by constraining the rate coefficients against the combustion data, as well as a consistency-screening process. Lastly, the uncertainty of the constrained model is calculated using an inverse spectral technique, and then propagated into a range of simulation conditions. The as-compiled model represents the prior model, while models constrained by global experiments are posterior models.

Uncertainty quantification in MUM-PCE is based on the stochastic spectral expansion method. The solution of a general dynamical model is expressed as a function of a vector of independent, identically distributed random variables,

$$u = u(\mathbf{y}, t, \boldsymbol{\xi}) \tag{2.6}$$

and expanded in a polynomial chaos basis. The expansion coefficients are obtained by projection using the expectation operator and orthogonality of the basis functions.

$$u(\mathbf{y}, t, \xi) = \sum_{i=0}^{\infty} u_i(\mathbf{y}, t) H_i(\xi) \tag{2.7}$$

To efficiently connect uncertainty quantification with chemical kinetic modeling, MUMPCE combines stochastic spectral expansion with solution mapping. Each uncertain rate parameter is normalized into a factorial variable,

$$x_i = \frac{\ln(k_i/k_{i,0})}{\ln f_i} \tag{2.8}$$

so that its nominal value is zero and its uncertainty is bounded between -1 and 1. Model responses are then expressed as a polynomial expansion in these variables,

$$g_r(\mathbf{x}) = g_{r,0} + \sum_{i=1}^{N} a_i x_i + \sum_{i=1}^{N} \sum_{j \geq i}^{N} b_{ij} x_i x_j + \cdots \tag{2.9}$$

with coefficients determined from sensitivity analysis or regression against computational experiments.

Uncertainty propagation is achieved by treating the factorial variables as random variables, which may be either uniformly or normally distributed. The uncertainty in a model prediction depends on the variance of the rate parameter but not strongly on the exact form of its distribution. For a sufficiently large number of active parameters, the resulting prediction uncertainty approaches a normal distribution by the Central Limit Theorem.

Model constraining is formulated as a least-squares optimization problem in which the difference between simulated responses and experimental observations is minimized subject to uncertainty bounds.

$$\Phi(\mathbf{x}^*) = \min_{\mathbf{x}} \sum_{r=1}^{n} \left(\frac{g_r(\mathbf{x}) - g_r^{\text{obs}}}{\sigma_r^{\text{obs}}} \right)^2 \tag{2.10}$$

A consistency analysis is introduced to evaluate whether experimental targets are internally consistent with the constrained model, using normalized scalar products and weighted residual measures to iteratively remove inconsistent targets.

Finally, uncertainty minimization is performed by interpreting the objective function as a joint probability density function for the rate parameters and deriving a covariance matrix for the posterior model. The reduction in uncertainty arises from strong coupling among rate parameters as the model becomes constrained by the experimental targets. Fundamental limitations remain due to the size, accuracy, and information content of available combustion data sets.

2.8.5 Model Discrepency

ralph c. smith book check the residuals of model v data for given mechanism for structure. If residuals are biased, model doesnt capture everything. solution 1: add an model discrepency term. fit it as a polynomial. solution 2 : (bayes land)

if model descrepency is due to shortcomings of experiments(i.e scneario dependent) we would expect it to work best when modeled as different functions for different QoIs. however if it is a shortcoming of the model(ie. missing reactions) a single function(e.g polynomial) fit or a single prior distrubition of the discrepency should be enought to partially compensate.?? at least me thinks thta way.

Shortcomings of previous work can be due to model discrepancies. Reaction mechanism does not contain all chemical reactions relevant to the combustion processs of interest. Because omitting reactions in the mecahnisms makes the computanations easier and and if there a re reactions that are not tknow to be in the combustion process it wouldn't be in there any ways. So we assume it is not possible to improve the model to compenstate for the discrepancy. Instead I will develeop a statistical method to try to partially identify and partially compensate this discrepancy. One such method

2.8.5.1 Consistency Check

First step of consistency analysis is to check for a given reaction model, whether its predictions fall within the uncertainties of experimental data.

2.8.6 Reaction Mechanism Optimization

Mechanism with reaction rates spanning different orders of magnitude are stiff.

2.8.7 wants/constraints

- should be integrated into TUMKIN. that means c++/rust. and reliable libraries - consistency analysis should be able to run with minimal human intervention. ideally no need to choose which reactions to optimize?

2.8.8 todo

- check the bib of mcmc paper

3 Methodology

Source code can be found at: https://github.com/barkinguney/consistency-analysis

3.1 Experimental Shock Tube IDT Uncertainty

Every dataset in TUMKIN needs to contain appropriate uncertainties. If authors don't report uncertainties for experimental data, uncertainties are calculated using the Uncertainty Quantification(UQ) module of TUMKIN. https://github.com/barkinguney/reaction-data-consistency-thesis

Reliable kinetic modeling and data-driven optimization require experimental databases with quantified uncertainties, yet more than 90% of published ignition-delay time (IDT) datasets report none, which limits machine learning and statistical calibration. This work presents an automated algorithm to reconstruct uncertainty for shock-tube (ST) ignition-delay times when it is not reported in the original publication.

The method follows GUM and NIST guidance and separates uncertainty into systematic (Type-B) and random (Type-A) components. The combined standard uncertainty is defined as the square root of the sum of the squared systematic and random contributions, and the expanded uncertainty is reported with a coverage factor k=2 , corresponding to approximately 95% confidence.

The systematic uncertainty captures device- and condition-specific biases in shock-tube IDT measurements and is treated as epistemic. Departures from an idealized zero-dimensional post-reflected-shock model are quantified by identifying dominant facility and operational non-idealities. Fourteen impacting factors describing hardware, operating conditions, diagnostics, and ignition-delay definition are selected to define proximity to an "ideal case." A baseline systematic error of 5% at 95% confidence is assigned to this quasi-ideal reference.

Deviations from the ideal metrics are mapped to incremental penalties using a modified Harrington desirability function, which retains low sensitivity near the ideal point and saturation for large deviations. The individual penalties are aggregated to yield the total systematic uncertainty. This approach operationalizes systematic uncertainty in cases where rigorous multi-laboratory statistical evaluation is not feasible.

Random uncertainty is obtained by propagating uncertainties in input parameters, such as reflected-shock temperature, pressure, and equivalence ratio, through an analytical ignition-delay correlation using the law of uncertainty propagation. Manufacturer specifications, calibration data, or reasonable estimates are used when uncertainties are not explicitly reported.

The workflow outputs combined and expanded uncertainties with standardized and traceable bounds suitable for database integration. The method is demonstrated on published shock-tube datasets, where expanded combined uncertainties between approximately 15% and 25% are obtained, with the largest relative uncertainties occurring for very short and very long ignition delays.

The proposed procedure enables standardized uncertainty reporting for shock-tube IDTs and supports database ingestion, data-consistency analysis, Bayesian parameter estimation, and mechanism optimization. The implementation is prepared for integration into the developing TUMKIN chemical-kinetics cyber-infrastructure at the Technical University of Munich.

3.1.1 Systematic Uncertainty

Systematic uncertainties are calculated by evaluating how much the experimental conditions deviate from ideal shock tube parameters. This is achieved using Harrington Desirability Functions, which transform nonidealities into a dimensionless penalty scale. 1.1 Penalty Calculation for Impacting Factors (IFs)For each impacting factor μ_{actual}, a penalty δ_i is calculated based on its proximity to an ideal range $[L, U]$. If the value falls within the range, the penalty is zero. If it falls outside, the penalty increases exponentially:

$$x_i = \frac{\mu_{ideal} - \mu_{actual}}{\mu_{ideal}}$$

$$\xi_i = 1 - \exp(-x_i^2)$$

$$\delta_i = \delta_0 \cdot \xi_i$$

Where δ_0 is a scaling constant (set to 5.0 in this study). For boolean factors (e.g., presence of impurities or tailoring), a fixed binary penalty is applied when the condition is not met.1.2 Total Systematic ErrorThe total systematic relative error for a specific data point, ε_{sys}, is the sum of penalties from fixed experimental factors and the specific ignition delay time (IDT) measurement:

$$\varepsilon_{sys} = \sum \delta_{fixed} + \delta_{IDT} + \delta_0$$

This relative error is then converted into an absolute systematic variance:

$$u_{sys}^2 = \left(\frac{\tau \cdot \varepsilon_{sys}}{100 \cdot 2}\right)^2$$

Category	#	Impacting Factor (IF)	Target metric, μ_i^0
Test equipment	1	Driver length	$\geq$ 3.0 m
	2	Driven length	$\geq$ 8.0 m
	3	Driven inner diameter	$\geq$ 10.0 cm
	4	Tailoring	yes
	5	Inserts	yes
	6	CRV	yes
Operating conditions	7	Reflected-shock Temperature, T_5	1000 - 1600 K
	8	Reflected-shock Pressure, P_5	5 - 20 atm
	9	Valid test time, Δt_{valid}*	50 - 500 µs
	10	Dilution (Ar/He/…)**	yes
	11	Impurities	no
	12	Measurement location	$\leq$ 2cm (from the end of the tube)
	13	Fuel-oxidizer equivalence ratio	$\geq$ 0.3
	14	Pre-shock pressures, P_1	$\geq$ 10 Torr

*Δt_{valid} time window before contact-surface/expansion-wave arrival or significant boundary-layer growth invalidates 0-D assumptions.
** "Yes" indicates purposeful dilution to suppress non-ideal gas dynamics and heat-release effects (typical practice is high inert fraction; specify per system if needed).

Figure 3.1: This is the description that appears under the image.

$$\begin{aligned}
\sigma_i^2(\tau_{ign}) = {} & \left(z_2^C z_3^D A \exp\left(\frac{B}{z_1}\right) \cdot \left(-\frac{B}{z_1^2}\right)\right)^2 \cdot u^2(z_1) + \left(CA \exp\left(\frac{B}{z_1}\right) z_2^{C-1} z_3^D\right)^2 \cdot u^2(z_2) \\
& + \left(DA \exp\left(\frac{B}{z_1}\right) z_2^C z_3^{D-1}\right)^2 \cdot u^2(z_3) + \left(\exp\left(\frac{B}{z_1}\right) z_2^C z_3^D\right)^2 \cdot u^2(A) \\
& + \left(\frac{1}{z_1} A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D\right)^2 \cdot u^2(B) + \left(A \exp\left(\frac{B}{z_1}\right) z_2^C \ln(z_2) z_3^D\right)^2 \cdot u^2(C) \\
& + \left(A \exp\left(\frac{B}{z_1}\right) z_2^C z_3^D \ln(z_3)\right)^2 \cdot u^2(D)
\end{aligned} \tag{3.1}$$

3.2 Arrhenius Parameters Uncertainty

Different sources report different values for the Arrhenius parameters or rate coefficients for the same reaction. Therefore, it is important to take all this information into account when analysing and optimizing reaction mechanisms. Kinetics databases like NIST chemical kinetics database and RMG database will be used to collect kinetics data, and calculate least squares fit to determine best parameters, and their distribution, specified by the covariance matrix of the least squares fit. Standard deviations of the parameters will be used to determine uncertainty bounds of the rate coefficients.

Workflow is as follows. For a given reaction equation string, reactant and product species are determined by comparing the string to known species. SMILES representations of the species are used to search RMG database libraries for matching sources. For every matching source, reaction rate coefficient is evaluated at fixed points in the valid temperature range of the source. In the RMG database most sources do not specify the valid temperature range of the reported kinetics data. In that case reaction rate coefficient is evaluate over a wide temperature range of 300K-3100K. Valid temperature ranges not being available for most sources is not ideal and is one of the downsides of the RMG database. Collected kinetics data, and rate coefficients evaluated at fixed intervals within the temperature range are saved further processing. Currently RMG kinetics types of Arrhenius, MultiArrhenius, ThirdBody are supported. For future development PDepArrhenius should be added. Currently efficencies of ThirdBody reactions are not considered.

In standard Arrhenius fitting parameters (A,n,E_a) are usually highly correlated.

$$\ln k(T) = \beta_0 + n \ln\left(\frac{T}{T_{ref}}\right) - \frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)$$

The rate expression is transformed into a log linear model centered around a reference temperature, which is geometric mean of temperature unless otherwise specified. The parameters (β_0, n, E_a) are solved using a Non-Linear Least Squares approach. The resulting covariance matrix in the reparameterized space, Σ_β, is then mapped back to the original $(\log_{10} A, n, E_a)$ space using the Jacobian transformation matrix J:

$$\Sigma_\theta = J \Sigma_\beta J^T$$

Diagonal entries of the covariance matrix of the least squares fit give parameter standard deviations $s\left(\log_{10}\left(A\right)\right), s\left(n\right), s\left(E_a\right)$, which are used to calculate lower and upper uncertainty bounds for rate coefficients.

$$k_{\text{low}}(T) = 10^{\log_{10}(A) - s(\log_{10} A)}\, T^{\,n-s(n)}\, \exp\left(-\frac{E_a + s(E_a)}{T}\right) \tag{3.2}$$

$$k_{\text{up}}(T) = 10^{\log_{10}(A) + s(\log_{10} A)}\, T^{\,n+s(n)}\, \exp\left(-\frac{E_a - s(E_a)}{T}\right) \tag{3.3}$$

The uncertainty parameter f is defined as:

$$f(T) = \log_{10}\left(\frac{k_{\text{up}}(T)}{k_0(T)}\right) = \log_{10}\left(\frac{k_0(T)}{k_{\text{low}}(T)}\right) \tag{3.4}$$

Where k_0 is the best fit of the rate coefficient, k_low and k_up are the lower and the upper bounds respectively [8]. This gives a temperature dependent uncertainty factor f(T). Temperature dependence of f(T) is usually low. To simplify further analysis, mean value of f(T) is selected as constant f. When rate coefficients need to be sampled in further anylysis, they are sampled from the interval $\left[k_0\left(T\right)/10^f\left(T\right), k_0\left(T\right) * 10^f\left(T\right)\right]$. In practice this can be achieved by multiplying the pre exponantial factor A witha multiplier sampled from $\left[\left(T\right)/10^f\left(T\right), \left(T\right) * 10^f\left(T\right)\right]$. Which preserves the shape of the k(T) curve and only scales it.

3.3 Cantera IDT Simulations

Cantera is a 0D and 1D chemical kinetics simulation software. To simulate shock tube IDT, constant volume batch reactors are used. Ignition delay times to be simulated for a given operating conditions T5, P5 phi and mechanism. The exact measurement or simulation time of IDT vary based on IDT definition. Using the same IDT definition as reported by the physical experiment to calculate simulated IDT would yield the closest results to physical experiment. the The ignition types that can be handled by this IDT simulation code are a follow. Following the format described by RESPECTH shock tube IDT measurements, ignition type is comprised of attributes, target, type, amount and unit . Target can be pressure, temperature or concentration. Type can be “max”, “d/dt max”, “baseline max/min intercept from d/dt“, “concentration”, “relative concentration”, and “relative increase”. ”max” means IDT is at the maximum of the target property. “d/dt max” means IDT is at the maximum gradient of target property. “baseline max/min intercept from d/dt“ means the time when the max d/dt line intercepts the baseline value of target property. “concentration”: IDT is at the time target property reaches a specified concentration value. “relative concentration”: IDT is at the time when the concentration of the target property relative to its maximum concentration reach the specified value. “relative increase”:IDT is at the time when the value of the target property increases by a given relative amount compared to its value at t=0. Amount attribute is only used when type are “concentration”, “relative concentration” or “relative increase”. Unit attribute is only used when the type is “concentration”, “relative concentration” or “relative increase”. Unit should have the value “unitless” for relative types.

3.4 Consistency analysis

Consistency methods, that check for a given dataset of experimental ignition delay time measurements and a mechanism with parameter bounds, whether experiments and mechanism are consistent with each other.

Figure 3.2: Flowchart.

3.4.1 Sensitivity analysis

For a given QoI, the model responses are sensitive to only a small subset of parameters. These parameters, referred to as active parameters, dominate the response behavior, a phenomenon referred to as effect sparsity [15]. Employing a sensitivity analysis method to identify these active parameters and using only these active parameters in further analysis improves computational feasibility. In [11] active parameters are identified by sampling the prior uncertainty region using space-filling designs like Latin hypercube sampling [16] or Sobol quasi-random sequence [17], and evaluating the original model at those points. A linear regression model Sl is then fitted to approximate the QoI as a function of the parameters.

$$S_l = \sum_{i=1}^{n} c_i x_i + c_0 \tag{3.5}$$

S_l contains information about how much the model output changes when each parameter is varied around its nominal value. The resulting regression coefficients define impact factors, which estimate how much the model output changes when each parameter varies across its uncertainty range. This is similar to sensitivity, but it also considers how large the parameter uncertainties are. A parameter with low sensitivity can still be high impact if it has high uncertainty.

$$\mathrm{IF}_i = |c_i|\ (u_i - l_i) = |c_i|\ \Delta x_i \tag{3.6}$$

Active parameters are selected by screening the ranked impact factors with a preset threshold [11]. Depending on the use case sensitivity analysis or uncertainty analysis, S_l or $\mathrm{I}Fi$ is used to rank parameters.

3.4.2 B2BDC Integration

3.5 Data Sources

When TUMKIN is up and running, uncertainty quantification and consistency analysis codes described in this thesis will use the data available in TUMKIN. The following sources has been used in the mean time to develop these modules, and could be imported into TUMKIN if suitable. Kinetics database RESPECTH [2] collect IDT, laminar flame speed, and concentration measurement s for various fuels. They are reported in their xml format. For testing the modules developed in this thesis I have used the RESPECTH data for shock tube IDT data for ethylene, Syngas and Hydrogen. Currently Most comprehensive collection of reaction rates and Arrhenius parameters is the NIST Kinetics Database [20]. That contain many literature sources for every reaction. However, they don't allow access to their database from code, no API or anything. It is the most comprehensive source, however, must be accessed manually. Reaction Mechanism Generator (RMG) on the other hand also contains a kinetics database that is openly accessible. For this thesis Arrhenius parameter data s taken from RMG database. To algorithmically extract kinetics data from RMG for a given reaction, SMILES or INCHI representations of the products and reactants are necessary. So, a species database that map between the widely accepted SMILES, INCHI representations and Cantera species name, and composition is necessary. An example format can be seen in OPTIMA++. Species names for the same species can be different across mechanism files if they are consistent within the mechanism. This is however not suitable for TUMKIN database. So before ingesting Mechanism files into TUMKIN it should be validated that species in mechanism can unambiguously be found in the TUMKIN species data base. This validating step is not in the scope of this thesis however is necessary for TUMKIN.

4 Results and Discussion

4.1 Bad Data

4.2 Good Data

4.3 Good Data + 1-2 Bad data

4.4 Experimental Uncertainty of Shock Tube IDT Data

4.4.1 shen vs davidson

4.4.2 Syngas

for most reported sources shock tube setup specific information is not available in an easily accessable database. They are available in the original papers,however to extract them would require manual entry, for programmatic processing with language model. This is necessary for an accurate estimation of experimental shock tube IDT uncertainties.

4.5 Cantera Shock Tube IDT Simulations

Current implementation of Cantera IDT simulation is tested with all available Syngas and Hydrogen IDT data from RESPETCH database. 4.4, 4.5, 4.6 show some example comparisons from that dataset.

4.6 illustrates the discrepency between constant volume reactor Cantera simulation and the real shock tube experiment, where boundary layer buildup reduces the effective volume. This

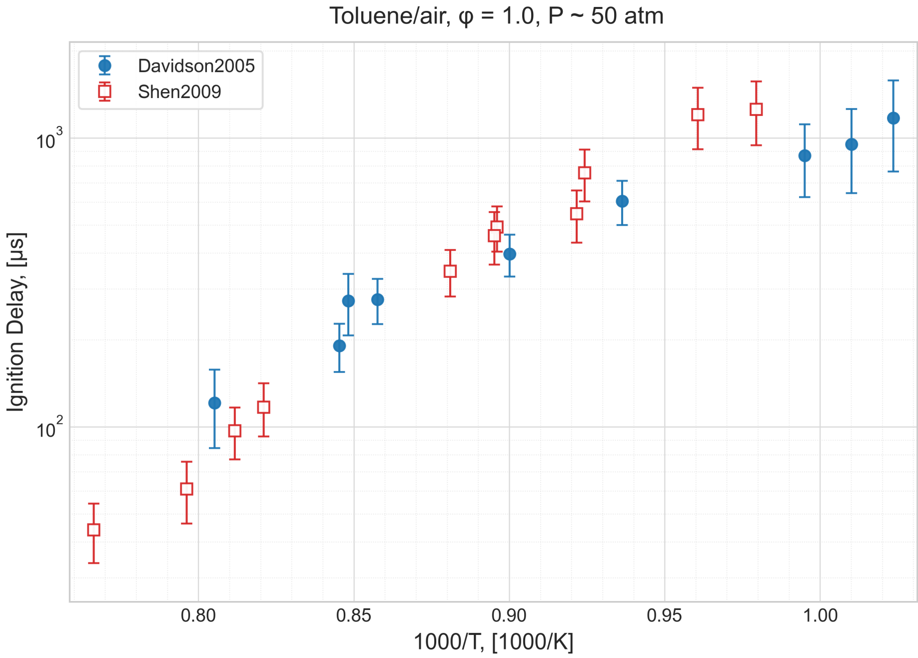

Figure 4.1: Comparison of calculated IDT uncertainties for a stoichiometric toluene/air mixture. IDT measurements are from two different facilities [18][19] over similar temperature range. P∼50 atm.

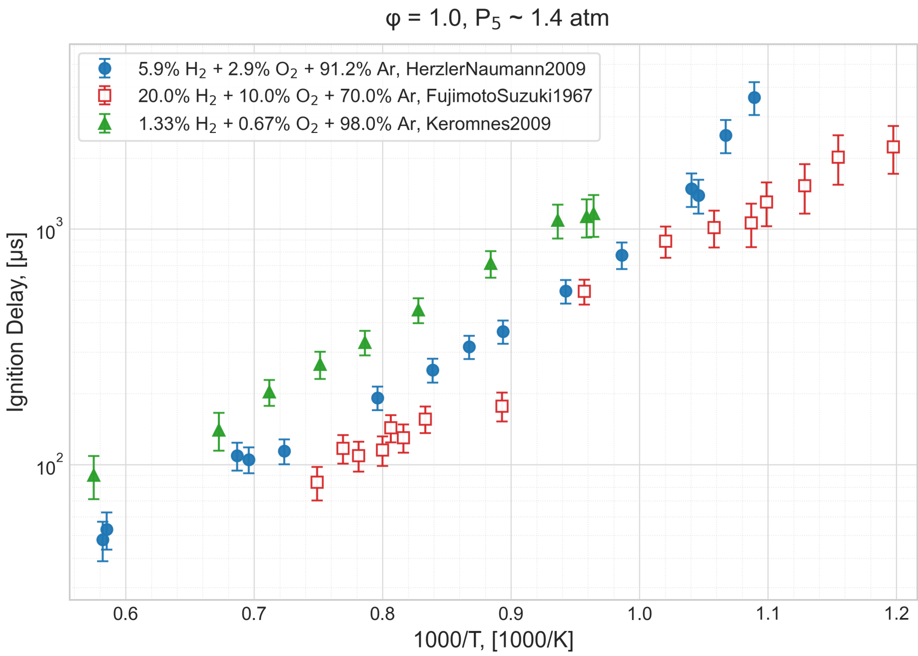

Figure 4.2: Comparison of calculated IDT uncertainties for $H_2/O_2/Ar$ mixtures. IDT measurements are from sources [20] [21] [22].$\phi = 1.0, P \sim 1.4 atm$.

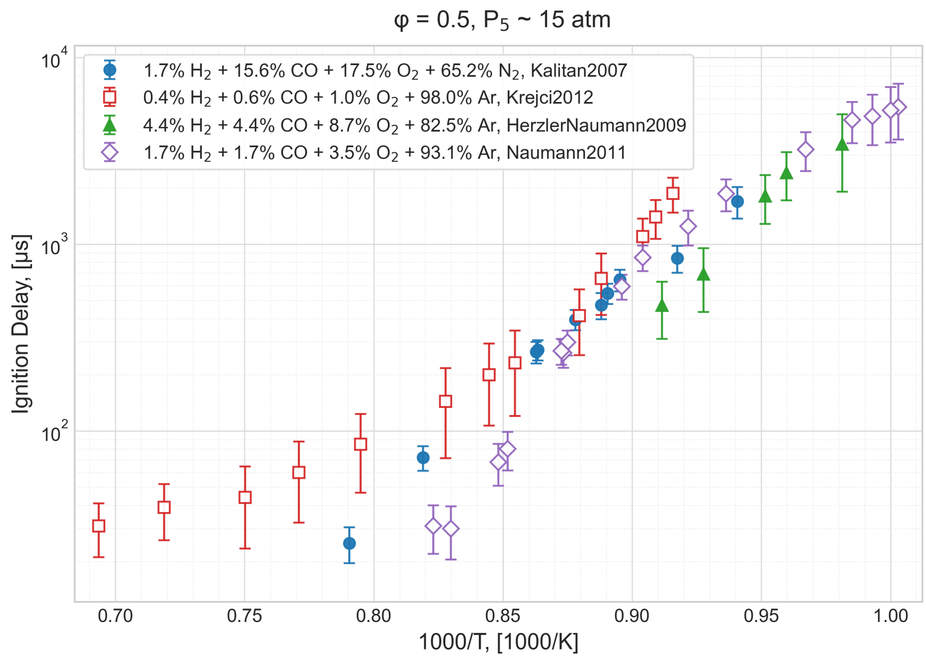

Figure 4.3: Comparison of calculated IDT uncertainties for diluted syngas/oxidizer mixtures. IDT measurements are from sources [23] [24] [20] [25]$\phi = 0.5, P \sim 15atm$.

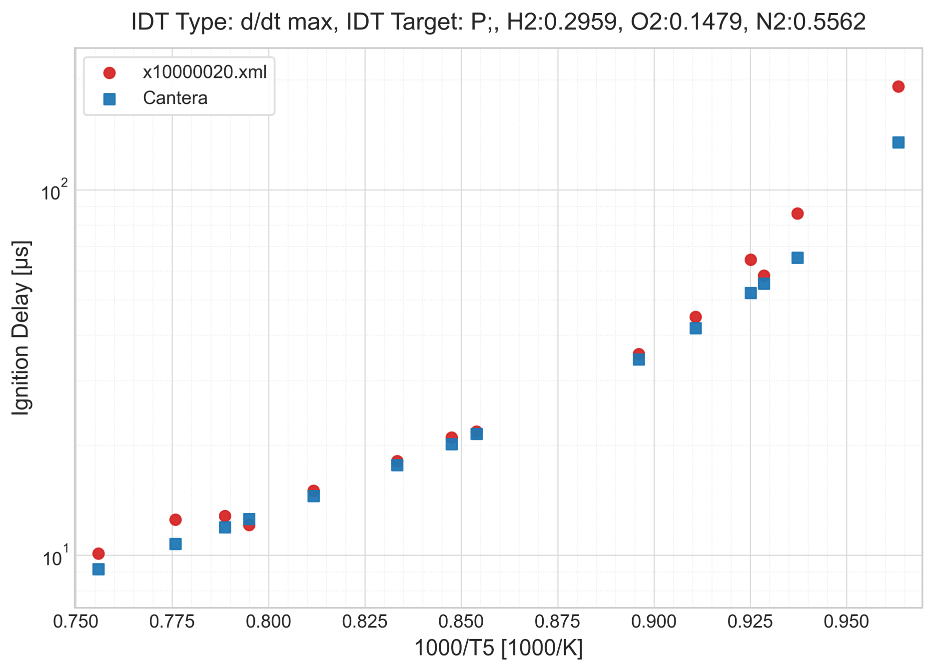

Figure 4.4: Comparison of Cantera IDT simulation with experimental measurement **good_cantera** IDT Type: dP/dt max.

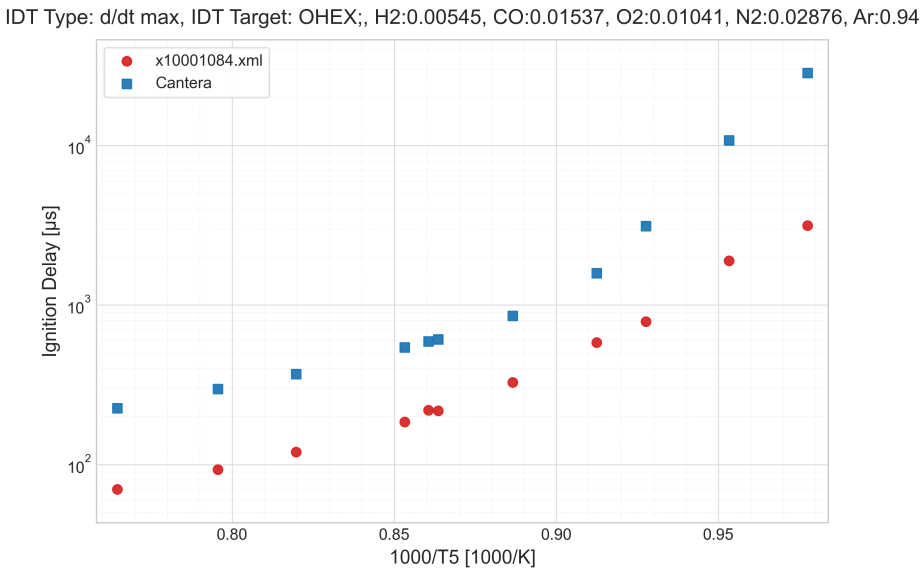

Figure 4.5: Comparison of Cantera IDT simulation with experimental measurement **good_cantera** IDT Type: d(OH^*)/dt max.

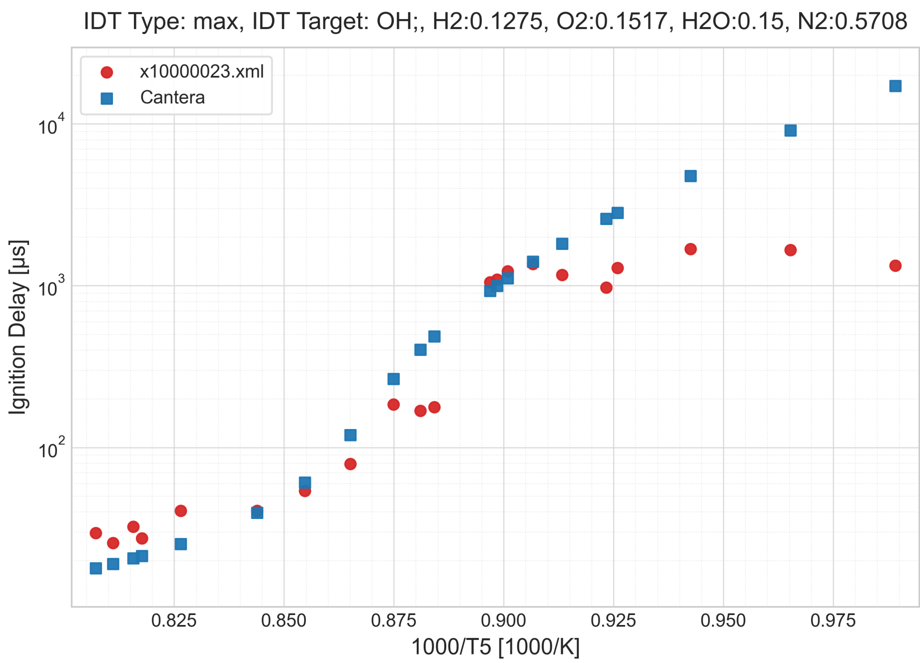

Figure 4.6: Comparison of Cantera IDT simulation with experimental measurement **good_cantera** IDT Type: OH max.

pressure rise should be compensated for in simulation to get more accurate IDT simulations. However in the RESPETCH syngas/hydrogen IDT datasets, pressure rise is rarely reported. It would be valuable for TUMKIN database to be more comprehensive.

4.6 Consistency Analysis

4.6.1 Sensitivity Analysis

4.6.2 Surrogate Model

4.6.3 B2BDC

4.6.3.1 Reaction Mechanisms

Reaction mechanisms used to validate the software developed in this thesis are Ethylene [18] and Combined Hydrogen Syngas [19]. Reported Chemkin input, transport and thermos files are converted to yaml by Cantera Chemkin to YAML conversion (c2kyaml).

5 Conclusion and Outlook

5.1 Outlook

A Appendix

B Program Flowcharts

References

C References

[1] M. Frenklach, "Transforming data into knowledge—Process Informatics for combustion chemistry," en, *Proceedings of the Combustion Institute*, vol. 31, no. 1, pp. 125–140, Jan. 2007, ISSN: 15407489. DOI: 10.1016/j.proci.2006.08.121. Accessed: Dec. 18, 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S1540748906003841.

[2] T. Turányi et al., "ReSpecTh: Reaction kinetics, spectroscopy, and thermochemical datasets," en, *Scientific Data*, vol. 12, no. 1, p. 1021, Jun. 2025, ISSN: 2052-4463. DOI: 10.1038/s41597-025-05272-6. Accessed: Dec. 25, 2025. [Online]. Available: https://www.nature.com/articles/s41597-025-05272-6.

[3] B. Su, M. Papp, I. Zsély, T. Nagy, P. Zhang, and T. Turányi, "Comparison of the performance of ethylene combustion mechanisms," en, *Combustion and Flame*, vol. 260, p. 113 201, Feb. 2024, ISSN: 00102180. DOI: 10.1016/j.combustflame.2023.113201. Accessed: Feb. 11, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0010218023005758.

[4] T. Varga et al., "Development of a Joint Hydrogen and Syngas Combustion Mechanism Based on an Optimization Approach," en, *International Journal of Chemical Kinetics*, vol. 48, no. 8, pp. 407–422, Aug. 2016, ISSN: 0538-8066, 1097-4601. DOI: 10.1002/kin.21006. Accessed: Feb. 11, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/kin.21006.

[5] T. Varga et al., "Optimization of a hydrogen combustion mechanism using both direct and indirect measurements," en, *Proceedings of the Combustion Institute*, vol. 35, no. 1, pp. 589–596, 2015, ISSN: 15407489. DOI: 10.1016/j.proci.2014.06.071. Accessed: Feb. 11, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S1540748914002296.

[6] M. S. Johnson et al., "RMG Database for Chemical Property Prediction," en, *Journal of Chemical Information and Modeling*, vol. 62, no. 20, pp. 4906–4915, Oct. 2022, ISSN: 1549-9596, 1549-960X. DOI: 10.1021/acs.jcim.2c00965. Accessed: Feb. 11, 2026. [Online]. Available: https://pubs.acs.org/doi/10.1021/acs.jcim.2c00965.

[7] H. Wang and D. A. Sheen, “Combustion kinetic model uncertainty quantification, propagation and minimization,” en, *Progress in Energy and Combustion Science*, vol. 47, pp. 1–31, Apr. 2015, ISSN: 03601285. DOI: 10.1016/j.pecs.2014.10.002. Accessed: Feb. 10, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0360128514000677.

[8] T. Nagy, É. Valkó, I. Sedyó, I. G. Zsély, M. J. Pilling, and T. Turányi, “Uncertainty of the rate parameters of several important elementary reactions of the H2 and syngas combustion systems,” en, *Combustion and Flame*, vol. 162, no. 5, pp. 2059–2076, May 2015, ISSN: 00102180. DOI: 10.1016/j.combustflame.2015.01.005. Accessed: Jan. 5, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0010218015000085.

[9] J. Prager, H. N. Najm, K. Sargsyan, C. Safta, and W. J. Pitz, “Uncertainty quantification of reaction mechanisms accounting for correlations introduced by rate rules and fitted Arrhenius parameters,” en, *Combustion and Flame*, vol. 160, no. 9, pp. 1583–1593, Sep. 2013, ISSN: 00102180. DOI: 10.1016/j.combustflame.2013.01.008. Accessed: Jan. 13, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0010218013000230.

[10] A. S. Tomlin, “The role of sensitivity and uncertainty analysis in combustion modelling,” en, *Proceedings of the Combustion Institute*, vol. 34, no. 1, pp. 159–176, 2013, ISSN: 15407489. DOI: 10.1016/j.proci.2012.07.043. Accessed: Feb. 10, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S1540748912003355.

[11] W. Li, “Development of Statistical and Deterministic Approaches to Uncertainty Quantification under Bound-to-Bound Data Collaboration,” Ph.D. dissertation. [Online]. Available: https://escholarship.org/uc/item/4ts9p04c.

[12] D. A. Sheen and H. Wang, “The method of uncertainty quantification and minimization using polynomial chaos expansions,” en, *Combustion and Flame*, vol. 158, no. 12, pp. 2358–2374, Dec. 2011, ISSN: 00102180. DOI: 10.1016/j.combustflame.2011.05.010. Accessed: Dec. 18, 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0010218011001568.

[13] S. Iavarone et al., “Application of Bound-to-Bound Data Collaboration approach for development and uncertainty quantification of a reduced char combustion model,” en, *Fuel*, vol. 232, pp. 769–779, Nov. 2018, ISSN: 00162361. DOI: 10.1016/j.fuel.2018.05.113. Accessed: Dec. 18, 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0016236118309360.

[14] A. Hegde, W. Li, J. Oreluk, A. Packard, and M. Frenklach, “Consistency Analysis for Massively Inconsistent Datasets in Bound-to-Bound Data Collaboration,” en, *SIAM/ASA Journal on Uncertainty Quantification*, vol. 6, no. 2, pp. 429–456, Jan.

2018, ISSN: 2166-2525. DOI: 10.1137/16M1110005. Accessed: Dec. 18, 2025. [Online]. Available: https://epubs.siam.org/doi/10.1137/16M1110005.

[15] M. Frenklach, H. Wang, and M. J. Rabinowitz, "Optimization and analysis of large chemical kinetic mechanisms using the solution mapping method—combustion of methane," en, *Progress in Energy and Combustion Science*, vol. 18, no. 1, pp. 47–73, Jan. 1992, ISSN: 03601285. DOI: 10.1016/0360-1285(92)90032-V. Accessed: Jan. 13, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/036012859290032V.

[16] M. D. McKay, R. J. Beckman, and W. J. Conover, "A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code," *Technometrics*, vol. 21, no. 2, p. 239, May 1979, ISSN: 00401706. DOI: 10.2307/1268522. Accessed: Jan. 13, 2026. [Online]. Available: https://www.jstor.org/stable/1268522?origin=crossref.

[17] I. Sobol', "On the distribution of points in a cube and the approximate evaluation of integrals," en, *USSR Computational Mathematics and Mathematical Physics*, vol. 7, no. 4, pp. 86–112, Jan. 1967, ISSN: 00415553. DOI: 10.1016/0041-5553(67)90144-9. Accessed: Jan. 13, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/0041555367901449.

[18] H.-P. S. Shen, J. Vanderover, and M. A. Oehlschlaeger, "A shock tube study of the autoignition of toluene/air mixtures at high pressures," en, *Proceedings of the Combustion Institute*, vol. 32, no. 1, pp. 165–172, 2009, ISSN: 15407489. DOI: 10.1016/j.proci.2008.05.004. Accessed: Dec. 18, 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S1540748908000126.

[19] D. Davidson, B. Gauthier, and R. Hanson, "Shock tube ignition measurements of iso-octane/air and toluene/air at high pressures," en, *Proceedings of the Combustion Institute*, vol. 30, no. 1, pp. 1175–1182, Jan. 2005, ISSN: 15407489. DOI: 10.1016/j.proci.2004.08.004. Accessed: Dec. 18, 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0082078404000566.

[20] J. Herzler and C. Naumann, "Shock-tube study of the ignition of methane/ethane/hydrogen mixtures with hydrogen contents from 0% to 100% at different pressures," en, *Proceedings of the Combustion Institute*, vol. 32, no. 1, pp. 213–220, 2009, ISSN: 15407489. DOI: 10.1016/j.proci.2008.07.034. Accessed: Feb. 22, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S1540748908003155.

[21] I. O. C. Chemical Kinetics Laboratory, *Fujimoto, S., Suzuki, M., Memoirs of the Defense Academy, Japan, 1967, (VII., 3), 1037-1046, Table 1.* 2017. DOI: 10.24388/X10000006. Accessed: Feb. 22, 2026. [Online]. Available: http://respecth.chem.elte.hu/respecth/reac/CombustionData.php?filedoi=10.24388/x10000006.

[22] A. Kéromnès et al., “An experimental and detailed chemical kinetic modeling study of hydrogen and syngas mixture oxidation at elevated pressures,” en, *Combustion and Flame*, vol. 160, no. 6, pp. 995–1011, Jun. 2013, ISSN: 00102180. DOI: 10.1016/j.combustflame.2013.01.001. Accessed: Feb. 22, 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0010218013000023.

[23] D. M. Kalitan, J. D. Mertens, M. W. Crofton, and E. L. Petersen, “Ignition and Oxidation of Lean CO/H2 Fuel Blends in Air,” en, *Journal of Propulsion and Power*, vol. 23, no. 6, pp. 1291–1301, Nov. 2007, ISSN: 0748-4658, 1533-3876. DOI: 10.2514/1.28123. Accessed: Feb. 22, 2026. [Online]. Available: https://arc.aiaa.org/doi/10.2514/1.28123.

[24] M. C. Krejci et al., “Laminar Flame Speed and Ignition Delay Time Data for the Kinetic Modeling of Hydrogen and Syngas Fuel Blends,” in *Volume 2: Combustion, Fuels and Emissions, Parts A and B*, Copenhagen, Denmark: American Society of Mechanical Engineers, Jun. 2012, pp. 931–948, ISBN: 978-0-7918-4468-7. DOI: 10.1115/GT2012-69290. Accessed: Feb. 22, 2026. [Online]. Available: https://asmedigitalcollection.asme.org/GT/proceedings/GT2012/44687/931/250546.

[25] I. O. C. Chemical Kinetics Laboratory, *C. Naumann; J. Herzler; P. Griebel; H. Curran; A. Keromnes; I. Mantzaras H2-IGCC project: Results of ignition delay times for hydrogen-rich and syngas fuel mixtures measured; 2011, Table 3.1, H2 / CO / O2 / Ar, Phi = 1.0, dilution 1:5, H2 /CO = 50/50, 16 bar series*, 2017. DOI: 10.24388/X10001024_X. Accessed: Feb. 22, 2026. [Online]. Available: http://respecth.chem.elte.hu/respecth/reac/CombustionData.php?filedoi=10.24388/x10001024_x.

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Moreover, I affirm that, so far, the thesis is not forwarded to a third party nor is it published. I obeyed all study regulations of the Technical University of Munich.

Munich, March 2, 2026

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